

**MASID: QUALITY MONITORING OF FLOOD CONTROL INFRASTRUCTURE
DELIVERY THROUGH PROCUREMENT AND SATELLITE DATA**

An Undergraduate Thesis Presented to
the Faculty of the College of Science
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Bachelor of Science in Mathematics with
Specialization in Computer Science

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CHAPTER III

METHODOLOGY

This chapter presents the methodology that will be used in the development and assessment of the proposed system. The study focuses on developing a data-driven framework for monitoring and objective verification of the physical realization of publicly funded flood control projects by combining government procurement data, multi-temporal satellite imagery, machine learning, Geographic Information Systems (GIS), and mathematical prioritization. The methodology covers the research design, study locale, sources of data, mathematical model and its adoption, process of developing the system, system evaluation, and ethical considerations. The procedures are designed to support the identification of project characteristics, analysis of satellite-observed changes, comparison between reported accomplishments and observable physical conditions, prioritization of projects for physical verification, and operationalization of the framework through a web-based GIS. This approach follows the intended MASID methodology of integrating project records, resolving project locations, observing sites through multi-temporal satellite imagery, detecting changes, and assessing agreement between reported and observed conditions.

Research Design

In this study utilizes a descriptive-developmental in the creation of MASID that integrates geospatial analysis, machine learning classification, and system development. The descriptive component of the study focuses on collecting, organizing, and analyzing appropriate information regarding flood control projects, such as procurement and contract details, geographic locations, reported accomplishments, implementation status, and observable physical conditions. On the other hand, the developmental component focuses on the design and

development of the MASID system, which is a GIS-based monitoring and visualization platform designed to implement the verification process.

The chosen design proceeds in strictly sequential phases aligned with the objective of the study. First, publicly funded flood control projects in the study area are profiled and geolocated to establish verified spatial footprints. Second, satellite-derived indicators are extracted, and machine learning models are trained and evaluated to estimate construction state and then classify delivery status. Third, officially reported accomplishments are systematically compared with satellite-observed evidence to quantify degrees of agreement or discrepancy. Fourth, a mathematical prioritization model is formulated to rank flagged projects for physical inspection on the basis of discrepancy severity and spatial risk factors. Finally, the MASID system is constructed and developed as a practical platform that presents verification results in support of full transparency and decision-making.

Research Locale

The study area is geographically delimited to the first district of Bulacan, Philippines. This district comprises Bulakan, Calumpit, City of Malolos, Hagonoy, Paombong, and Pulilan. The background of the study identifies Bulacan as one of the areas that received significant attention in investigations involving reported flood control projects, including concerns involving ghost, partially constructed, and substandard projects. By concentrating on a single congressional district rather than attempting nationwide coverage, the study guarantees a small yet manageable data volume while still capturing a meaningful concentration of flood control infrastructure projects that have been subject to public investigation. Thus, the findings are intended to demonstrate a real, scalable verification approach instead of producing unrealistic verification nationwide.

Sources of Data

The first major source of data is the government procurement and project records. These records will provide administrative and project-related information such as project identification, procurement and contract details, project location, implementation schedule, project status,

and reported accomplishments. The study specifically intends to cross-reference procurement records with multi-temporal satellite imagery to independently verify whether the reported project implementation corresponds with observable physical conditions. Such records are obtained from publicly available disclosures, request documents, and other implementing agencies, specifically the Department of Public Works and Highways.

The second major source is the multi-temporal satellite imagery. Satellite images provide spatial evidence of physical changes occurring at the identified project locations over different periods. The study identifies satellite missions such as Sentinel and Copernicus as potential sources of Earth Observation data. These images will be used to examine construction-related changes and estimate the construction state and delivery status of flood control infrastructure. The third major source is geospatial and geographic data. These data support the geolocation of infrastructure projects and the GIS-based analysis of their spatial relationships and other surrounding conditions. This can be used to integrate procurement information then geographic locations and satellite observations in order to visualize projects, analyze spatial patterns, and compare reported project information with satellite-observed conditions.

The last major source is the relevant environmental and spatial datasets if necessary for the analyzation on spatial risk factors and project prioritization. The GIScience foundation of the study recognizes the integration of different datasets, and these include engineering documents, satellite images, procurement database, environmental records, and field observations, within a common geospatial environment of the First District of Bulacan. Altogether, these sources allow the study to cross-reference factual claims with independent geospatial evidence and, in doing so, address the core verification gap that the MASID framework seeks to close.

Mathematical Model

The mathematical model developed in this study applies the Travelling Salesman Problem (TSP) to determine the most time-efficient and practical route for conducting physical inspections of flood control infrastructure projects that have been flagged by the MASID verification pipeline. Once the projects within the First District of Bulacan have been identified through the integration of procurement records, multi-temporal satellite imagery, machine learn-

ing, classification, and GIS-based spatial analysis, the model generates an optimized inspection sequence. The objective function is to minimize the total travel distance while ensuring that every flagged project is visited exactly once and that the route remains operationally feasible for field teams.

Table 1: *List of Mathematical Notation and Description*

<i>Notation</i>	<i>Description</i>
P_i	Priority score of project i
w_1	Weight for discrepancy
w_2	Weight for spatial risk
D_i	Discrepancy score
R_i	Spatial risk score
F	Set of all locations in the inspection route
m	Total number of flagged projects included in the route
i	Origin or the current location of the inspection
j	Destination or the next location of the inspection
d_{ij}	Travel distance from location i to location j
x_{ij}	Decision whether to travel from location i to location j
Z	Total travel distance
u_i	Position or the order of location i in the route
u_j	Position or the order of location j in the route
n	Number of project locations to be inspected

Priority Score

The model first undergoes prioritization for each flagged project for route optimization. The priority score may be calculated using the severity of the discrepancy between the reported project accomplishment and the satellite imagery, including relevant spatial risk factors.

$$P_i = w_1 D_i + w_2 R_i$$

A higher P_i means that the project has a higher priority for physical inspection. D_i is the discrepancy score that represents the degree of difference between the reported project accomplishment and the observed condition from satellite-based analysis. R_i is the spatial risk score, and w_1 and w_2 are their corresponding weights, such that $w_1 + w_2 = 1$ which means that the weights collectively represent a full 100% of the priority score.

Infrastructure projects with higher priority scores are considered to be picked first when selecting projects for physical inspection. After that, the geographic locations are readily provided to the TSP model to determine the shortest inspection route. The resulting route provides an efficient sequence with less time spent visiting the selected flood control infrastructure projects while minimizing travel distance.

Set of Locations

$$F = \{0, 1, 2, 3, \dots, m\}$$

The set of finite inspection locations of the projects is represented by F , where 0 represents the initial point of the inspection base, while 1 through m represent the total number of flagged flood control project locations. This determines the possible locations to be visited and the order in which they are included in the inspection route.

Travel Distance

For each pair of locations i and j , let d_{ij} be the travel distance covered from location i to location j , which is gathered from the geographic coordinates and road network data through GIS.

Decision Variable

A decision variable x_{ij} is defined as

$$x_{ij} = \begin{cases} 1, & \text{if the inspection route travels from location } i \text{ to location } j, \\ 0, & \text{otherwise.} \end{cases} \quad \text{for all } i, j \in F, i \neq j.$$

The variable x_{ij} tells the mathematical model whether a particular connection from i to j is included in the inspection route. A value of 1 indicates that the connection is selected, while a value of 0 indicates that the connection is not selected. Because x_{ij} can only take the values 0 or 1, it is called a binary decision variable.

Objective Function

The objective of the TSP model is to minimize the total travel distance.

$$\min Z = \sum_{i=0}^n \sum_{\substack{j=0 \\ j \neq i}}^n d_{ij} x_{ij}$$

The purpose of this minimization is to find the route with the smallest possible total travel distance. The TSP chooses the route with the shortest total distance.

Constraint 1 – Every Location Must Be Entered Exactly Once

For every project location j , exactly one incoming route must exist. This constraint is expressed as:

$$\sum_{\substack{i=0 \\ i \neq j}}^n x_{ij} = 1, \quad \text{for all } j \in F$$

It guarantees that no location is skipped and that no location is visited more than once from different preceding sites. Without this restriction, the solution could omit critical flagged projects or produce invalid, incomplete inspection paths.

Constraint 2 – Every Location Must Be Left Exactly Once

Symmetrically, every project location i must have exactly one outgoing route:

$$\sum_{\substack{j=0 \\ j \neq i}}^n x_{ij} = 1, \quad \text{for all } i \in F$$

This ensures that the inspection team does not remain indefinitely at any site and that the route continues systematically until all flagged projects have been covered. Together with Constraint 1, these degree constraints enforce that each project is visited precisely once.

Subtour Elimination Constraint

Subtour elimination constraints are used to make sure that the resulting output forms one complete inspection route rather than several disconnected smaller routes.

$$u_i - u_j + nx_{ij} \leq n - 1, \quad i \neq j$$

The u_i and u_j variables help the model determine the order in which projects are visited. Thus, the constraint prevents the TSP from forming disconnected smaller tours.

Process of Developing the System

The study utilizes the Agile methodology as the primary software development approach for MASID. Agile emphasizes iterative development, continuous refinement, collaboration, and responsiveness to changing requirements. This approach is appropriate for the study because the development of MASID is dependent on the availability, completeness, spatial characteristics, temporal coverage, and quality of procurement and satellite datasets. These characteristics may only become fully understood as the researchers acquire, prepare, and integrate the datasets.

Rather than developing the entire system in a single development cycle, the researchers will progressively implement the system through a series of iterations or sprints. Each iteration will involve planning, implementation, testing, review, and refinement. System components developed during an iteration will be assessed against the identified requirements before subsequent components are implemented. Issues concerning data structure, project geolocation, satellite imagery, system integration, visualization, and usability will be documented and addressed during succeeding iterations.

The Agile process will be applied to the technical development of the system, while the thesis project itself will continue to follow the academic requirements and schedule established by the researchers and the university. Initial consultation with relevant government agencies, particularly the Department of Public Works and Highways (DPWH) and the Philippine Space Agency (PhilSA), will provide domain-specific and data-related information that may inform the feasibility and design of MASID.

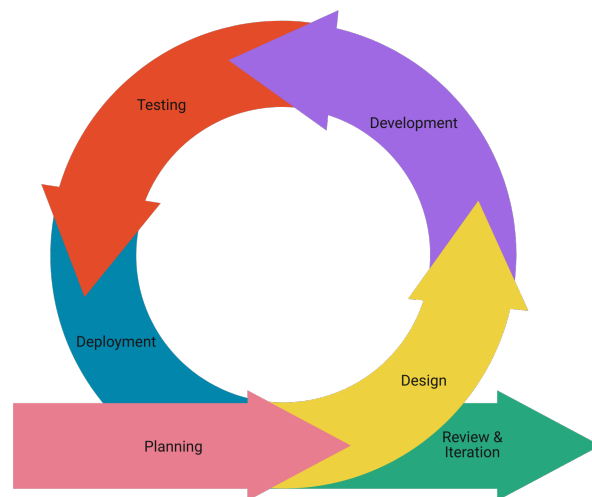


Figure 1: *Agile Development Cycle for MASID*

Planning

In this phase, the researchers established the scope, objectives, requirements, and overall direction of MASID: Quality Monitoring of Flood Control Infrastructure Delivery Through Procurement and Satellite Data. The planning process identified the key system features, data requirements, and functionalities necessary for monitoring flood control infrastructure delivery through the integration of procurement and satellite-based information. It also involved identifying relevant government agencies and potential data sources that could support the study. Initial consultations with the Department of Public Works and Highways (DPWH) and the Philippine Space Agency (PhilSA) were considered in determining the availability, relevance, and appropriate use of procurement and satellite data for the proposed system.

Additionally, this phase includes the thesis project plan, which outlines the timeline and major activities of the study. The Gantt chart presents the phases of the thesis project, including

Thesis One Output, Thesis Two Output, Data Collection and Preparation, Paper Miscellaneous, System Design, System Development, and System Assessment. Each phase addresses specific objectives and deliverables that contribute to the completion of the MASID system.

Gantt Chart

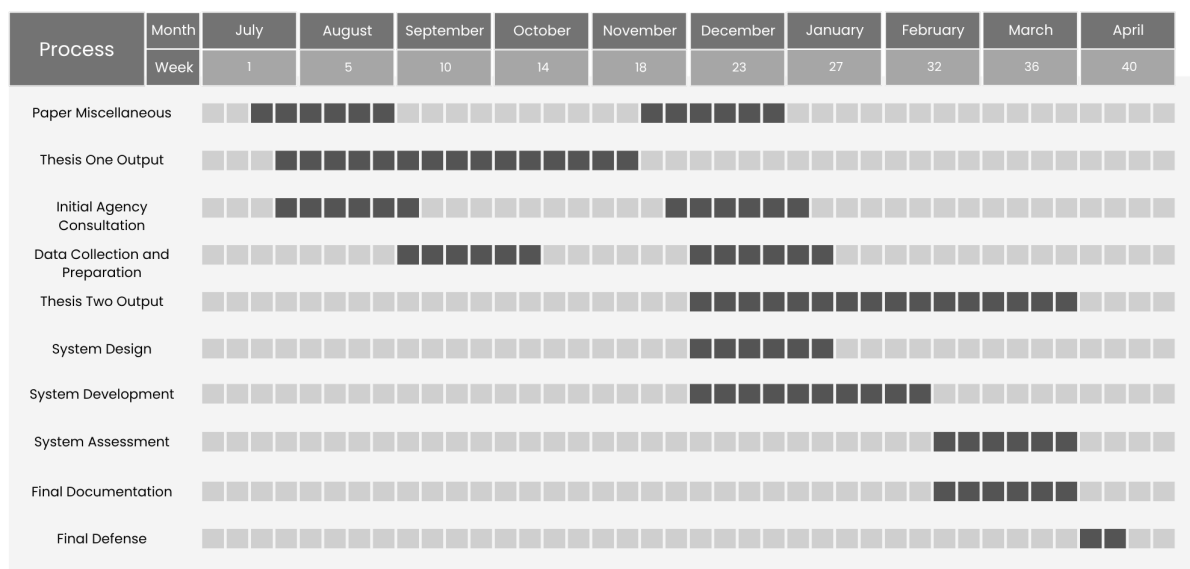


Figure 2: Gantt Chart

Thesis One Output

This phase involves developing the foundational components of the thesis, including the Introduction, Review of Related Literature, and Research Methodology. It covers the formulation of the study’s rationale, objectives, scope and delimitations, significance, and methodological approach. The researchers also prepare the proposal manuscript, presentation slides, and script for the mock defense and proposal defense.

Alongside the preparation of the thesis manuscript, this phase includes the initial consultation and coordination with relevant government agencies, particularly the Department of Public Works and Highways (DPWH) and the Philippine Space Agency (PhilSA). These consultations help the researchers understand the availability and nature of relevant procurement, flood control project, satellite, and geospatial data, as well as obtain domain-specific insights

that inform the feasibility and direction of the proposed system. The researchers also organize relevant literature through a literature review matrix to establish and support the contents of Chapter 2.

Thesis Two Output

Following the foundational work completed during the first thesis phase, this phase focuses on completing the remaining sections of the thesis, particularly the presentation and discussion of results, conclusions, and recommendations. It also includes the preparation of the final thesis manuscript, mock defense, and final thesis defense. This phase signifies the completion and finalization of the research study.

Requirements Analysis

During the Requirements Analysis phase, the researchers identify and analyze the functional, non-functional, data, and system requirements of MASID. Requirements will be derived from the objectives of the study, related literature, existing monitoring and geospatial systems, available datasets, initial agency consultations, and technical considerations identified during development.

The requirements analysis will determine the information that the system must process, the functions that users must be able to perform, and the quality characteristics that the system must satisfy. Particular attention will be given to the integration of procurement records with project locations and satellite-based information. Because the availability and structure of datasets may change as data collection progresses, identified requirements may be refined throughout the Agile development process.

Table 2: *Preliminary Functional Requirements of MASID*

ID	Functional Requirement	Description
FR-01	Project Visualization	Display available flood control projects geographically.
FR-02	Project Selection	Allow users to select a project from the map.
FR-03	Project Information	Display available information associated with a selected project.
FR-04	Procurement Information	Present relevant procurement records associated with a project.
FR-05	Geospatial Information	Present project location and geographic attributes.
FR-06	Satellite Information	Present available satellite-based information corresponding to a project.
FR-07	Temporal Comparison	Allow users to examine available information across different periods.
FR-08	Data Filtering	Allow users to narrow displayed projects based on available attributes.
FR-09	Search	Allow users to locate available projects or records.
FR-10	Integrated Project View	Present procurement, project, geographic, and satellite information together.

Table 3: Preliminary Non-Functional Requirements

Category	Requirement
Usability	The interface should allow users to navigate and examine project information with minimal unnecessary steps.
Functional Suitability	Implemented functions should correspond to the established system requirements.
Reliability	The system should provide consistent results when the same valid data and operations are used.
Performance	Map, project information, and available imagery should load within reasonable response times under the test environment.
Compatibility	The web application should operate on supported modern web browsers.
Security	Access to system resources and stored information should be appropriately controlled.
Maintainability	System components should be organized to facilitate future modification.
Data Integrity	Information presented by the system should preserve the relationships and values contained in the prepared datasets.

Data Collection and Preparation

This phase involves collecting and preparing the datasets required for the development, analysis, and evaluation of MASID. The researchers will coordinate with relevant government agencies and data providers to obtain procurement and flood control project records, satellite imagery, and supporting geospatial information.

Formal data requests and necessary documentation will be prepared where applicable. The researchers will document the source, format, date, coverage, and relevant attributes of each dataset. The collected datasets will undergo inspection, cleaning, transformation, and preparation before being integrated into the system.

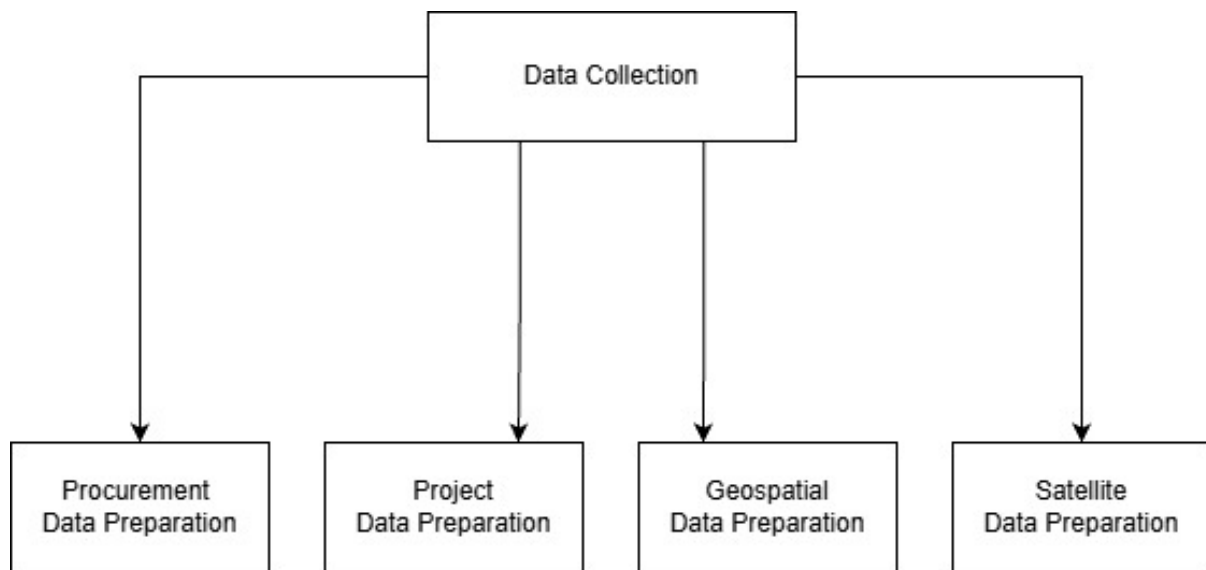


Figure 3: *MASID Data Preparation Pipeline*

Procurement Data Preparation

Procurement records will be examined to identify information relevant to flood control infrastructure projects. Depending on the fields available in the obtained dataset, relevant attributes may include project identification, project description, procuring entity, procurement status, contract information, project cost, dates, and other available procurement-related attributes.

The researchers will standardize field names and data formats, identify duplicate records, inspect missing values, and establish relationships between procurement records and individual project records. Fields that cannot be reliably associated with a specific project will be documented rather than assigned through unsupported assumptions.

Project Data Preparation

Flood control project records will be organized into a standardized project dataset. Each project will be assigned a unique internal identifier to facilitate the association of procurement information, geographic information, and satellite observations. Project names, descriptions, locations, dates, and other available attributes will be standardized to minimize inconsistencies across data sources.

Geospatial Data Preparation

Geospatial data will be prepared to establish the geographic representation of flood control projects. Coordinates and other location information will be validated and transformed into a consistent geographic reference system where necessary. Records with incomplete or uncertain geographic information will be flagged and documented.

Satellite Data Preparation

Available satellite imagery will be collected from appropriate sources based on the coverage, temporal requirements, spatial characteristics, and accessibility applicable to the study. The researchers will document the source, acquisition date, spatial resolution, geographic coverage, and other relevant metadata of each image.

Where applicable, satellite imagery will undergo preprocessing procedures required by the selected imagery and analytical method. These procedures may include image selection, cloud screening, clipping to the project area, coordinate-reference alignment, and other preprocessing operations required to ensure consistency between imagery and project locations.

The availability and quality of satellite imagery will be treated as methodological considerations. Missing imagery, cloud-covered observations, insufficient spatial resolution, or inadequate temporal coverage will not automatically be interpreted as evidence of non-delivery or project failure.

Data Integration

The prepared datasets will be integrated by establishing relationships between project identifiers, procurement records, geographic information, and satellite observations. The primary relationship will be established at the project level, where available.

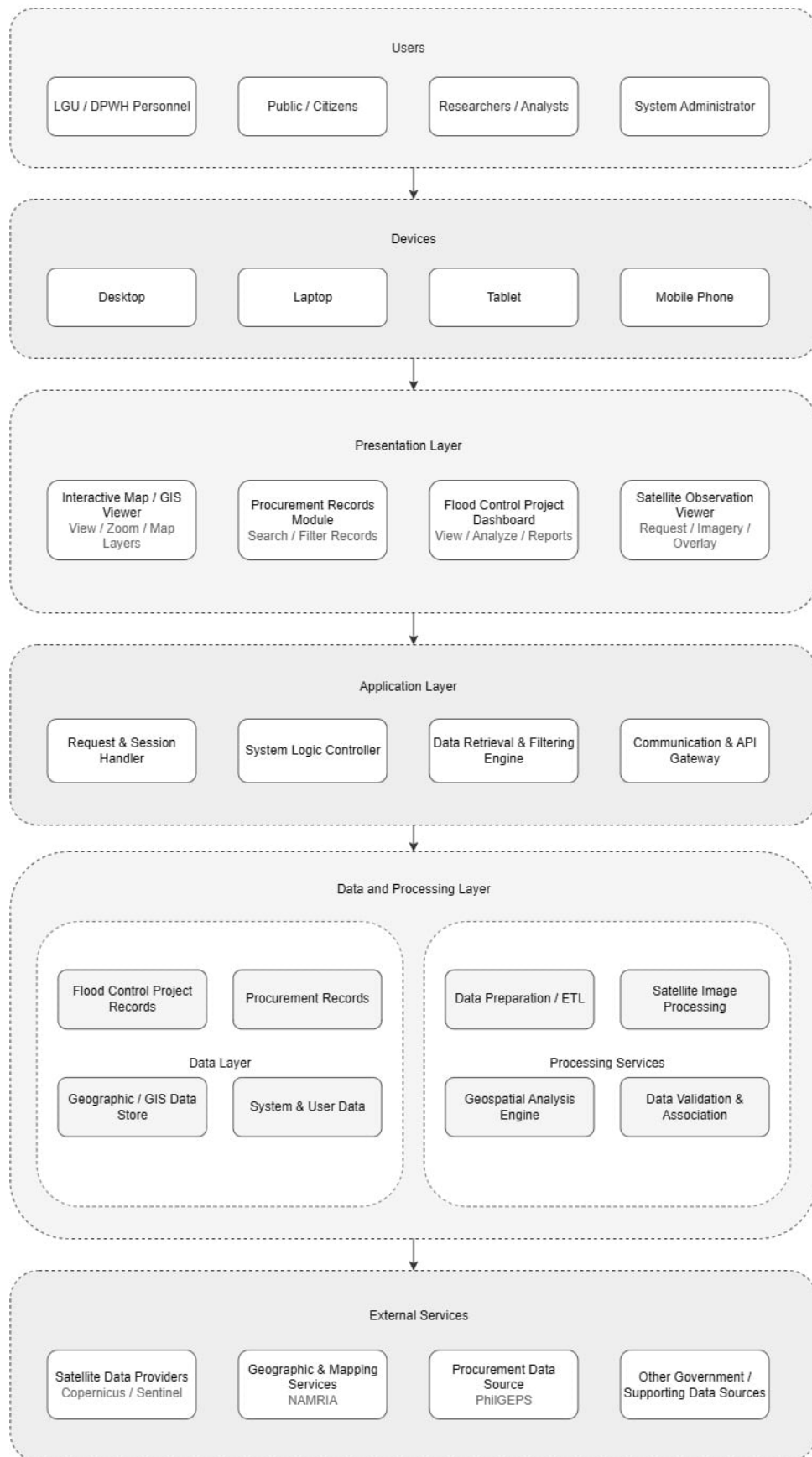


Figure 4: Data Integration Model

System Design

During the Design Stage of Agile, the researchers will convert the identified requirements into a formal technical and functional plan for MASID. The design will define the system architecture, database structure, data flow, process flow, geospatial interface, satellite-data integration, visualization components, and user interactions.

The design will establish how procurement records, flood control project information, geographic information, and satellite observations will be represented and associated within the system. Interface mockups, system architecture diagrams, entity-relationship diagrams, flowcharts, and other design specifications will be prepared to guide implementation.

System Architecture and Integration Design

MASID will utilize a layered web-based architecture consisting of a presentation layer, application layer, data and processing layer, and external data services. The architecture is designed to separate user interaction from application logic, data storage, and computational processing.

The presentation layer will provide the web-based interface through which users interact with the map, project records, procurement information, and available satellite observations. The application layer will manage requests, system logic, data retrieval, filtering, and communication between the interface and backend services. The data layer will store structured project, procurement, geographic, and system information. Processing services will perform the required data preparation and satellite-related computations. External data services will provide access to appropriate satellite, geographic, or other supporting information where applicable.

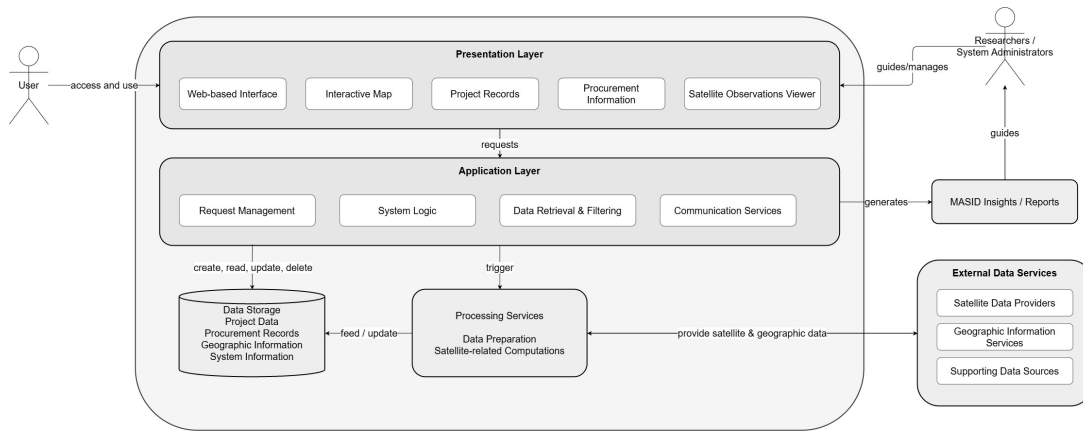


Figure 5: Proposed MASID System Architecture

Database Design

A relational database will be used to organize the structured information required by MASID. The database will maintain relationships between projects and their associated procurement, geographic, and satellite-related records.



Figure 6: Draft Entity-Relationship Diagram

The final database structure will be modified according to the actual datasets obtained and the final requirements of the system.

System Process Flow

Following the development of the system architecture, flowcharts will be developed to illustrate the sequential operations of MASID. The overall process begins with the acquisition of procurement, project, satellite, and geospatial datasets. The datasets are inspected, cleaned, standardized, and associated with the appropriate project records. Geographic information is then used to establish the project location within the system, while available satellite observations are associated with the corresponding project area and observation period.

The resulting integrated information is stored and made available through the web interface. Users can locate projects through the map, select a project, examine its available procurement and project information, and access corresponding satellite-based information. Where multiple observations are available, users can compare information across different periods.

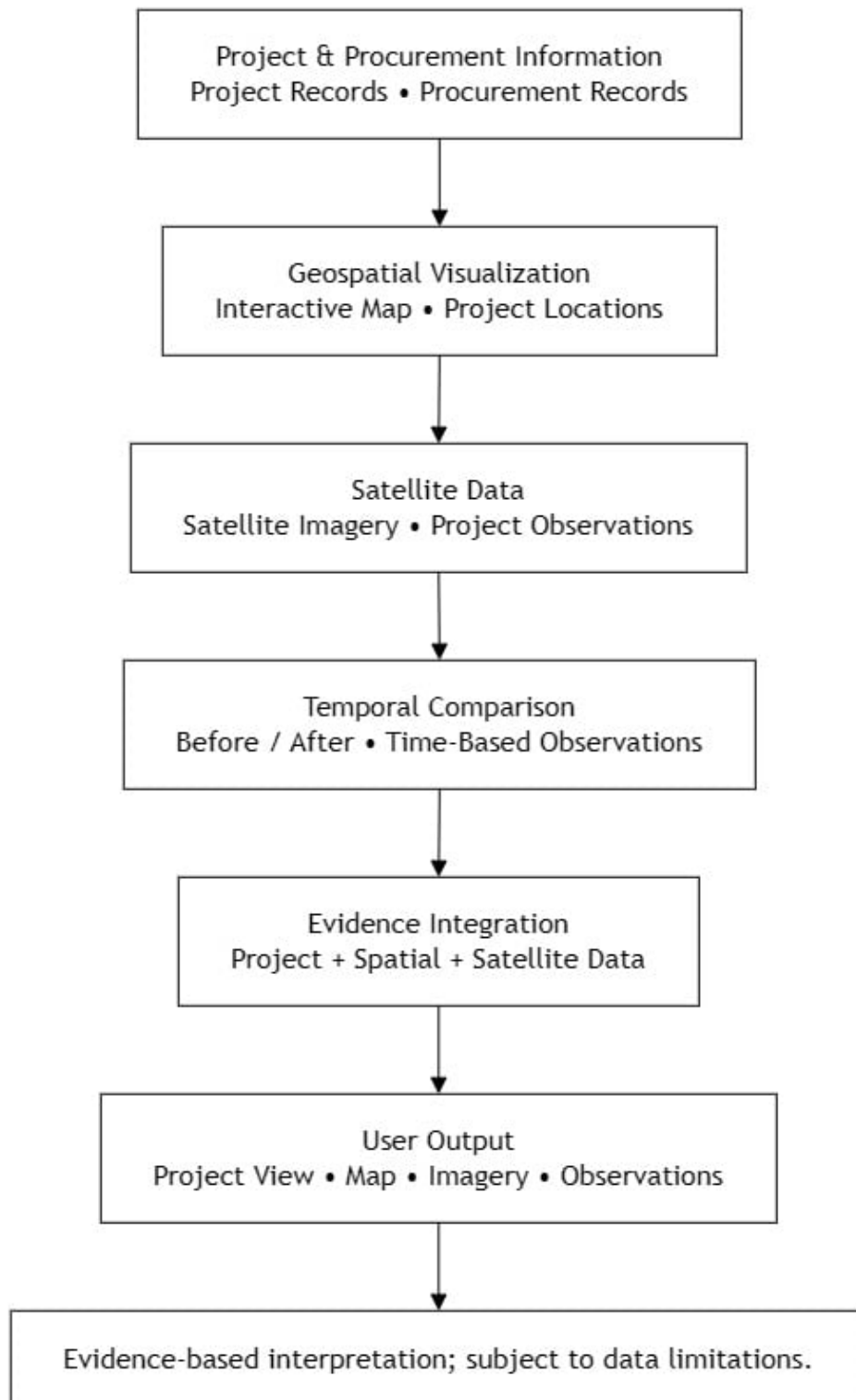


Figure 7: *Overall MASID Process Flow*

Project and Procurement Information Module

The Project and Procurement Information Module will organize and display the available information associated with individual flood control infrastructure projects. Each project will be represented through a unique project identifier to facilitate the association of project information with procurement and spatial records.

When a user selects a project, the system will retrieve the corresponding project and procurement records from the database. Depending on the availability of information, the project view may include project description, location, procurement information, contract-related details, project dates, status, and other relevant attributes contained within the source dataset.

The module will preserve the distinction between recorded procurement information and observations obtained from satellite imagery. Procurement information will represent information contained in the source records, while satellite-based information will provide spatial or temporal evidence that can be examined alongside the recorded project information.

Geospatial Visualization Module

The Geospatial Visualization Module will serve as the primary geographic interface of MASID. Flood control projects with sufficient location information will be represented as identifiable features on an interactive map. Users will be able to navigate the map, identify available projects, and select individual project markers to retrieve additional information.

The geographic interface will connect each project location to its corresponding project record. Geographic information may include coordinates, administrative location, project area, and other spatial attributes available in the prepared dataset. Records with incomplete or uncertain geographic information will be appropriately flagged rather than assigned unsupported locations.

Satellite Data Module

The Satellite Data Module will incorporate available satellite imagery corresponding to flood control project locations and observation periods. Satellite imagery will be processed

according to the selected analytical procedure and associated with the relevant project record.

Where multiple satellite observations are available, the system will organize the observations according to their acquisition dates. Users will be able to access available imagery or processed satellite-based information alongside the corresponding project and procurement records.

The module will explicitly account for limitations associated with satellite observations, including spatial resolution, cloud cover, image availability, temporal gaps, and other image-quality considerations. Such limitations will be documented and considered during interpretation.

Temporal Comparison Module

The Temporal Comparison Module will allow users to examine available project-related information across different periods. The interface may provide a temporal slider or equivalent date-selection mechanism through which users can navigate available satellite observations and corresponding project information.

The module is designed to facilitate visual examination of observable spatial changes over time. It will not automatically classify a project as successful, failed, completed, or defective solely on the basis of differences or similarities between satellite observations. Interpretation will remain dependent on the available evidence and the limitations of the underlying data.

User Interface Design

The user interface will be designed to provide a clear and organized means of examining flood control project information. The main interface will provide access to the interactive map and available projects. Users will be able to select a project to retrieve its corresponding project, procurement, geographic, and satellite information.

The interface will prioritize readability, navigation, and clear separation of information sources. Visual elements will be designed to distinguish recorded project information from satellite-based observations. Filtering and search functions will be incorporated where required

by the finalized system requirements.

Development

During the Development Stage of Agile, the researchers will implement, integrate, test, and refine the components established during the Design Stage. Development will be performed through successive iterations or sprints. Each sprint will produce a functional increment of MASID that can be tested and reviewed before succeeding components are implemented.

Development activities will cover the database, data processing components, web interface, mapping functions, procurement integration, satellite-data integration, temporal comparison, and system integration. The exact implementation priorities may be adjusted according to dataset availability and issues identified during preceding iterations.

Proposed Sprint Structure

Table 4: *Proposed Agile Sprint Plan*

Sprint	Primary Focus	Major Activities	Expected Output
Sprint 1	Foundation and Data	Database structure, dataset organization, project identifiers, initial interface	Working system foundation
Sprint 2	GIS and Mapping	Interactive map, project markers, geographic data	Functional project map
Sprint 3	Satellite Integration	Image integration, project-image association, temporal organization	Satellite-enabled project view
Sprint 4	System Integration	Procurement + project + GIS + satellite integration	Integrated MASID prototype
Sprint 5	Refinement	Testing, debugging, interface refinement, evaluation preparation	Evaluation-ready prototype

These sprint assignments should be finalized against the actual work schedule of the researchers before inclusion in the final manuscript.

Sprint 1: System Foundation and Data Integration

The first sprint will focus on establishing the technical foundation of MASID. The researchers will establish the initial project database structure, define project identifiers, prepare the data model, and integrate the initial prepared datasets. The basic web application structure will also be established during this iteration.

Initial validation will be performed to ensure that project records can be retrieved from the database and that relevant attributes are stored consistently. Problems involving missing fields, inconsistent identifiers, duplicate records, and incompatible data formats will be documented and addressed.

Sprint 2: Geospatial Visualization

The second sprint will focus on the implementation of the geospatial component of MASID. The researchers will integrate geographic information into the application and develop the interactive map interface. Project markers will be displayed according to the validated geographic information contained in the prepared dataset.

Testing will verify that project identifiers correspond to the correct geographic records and that users can select mapped projects to retrieve the corresponding project information.

Sprint 3: Satellite Data Integration

The third sprint will focus on integrating available satellite-based information into the system. Satellite observations will be organized according to project location and observation date. The researchers will implement the necessary data-processing and retrieval mechanisms required to associate satellite observations with corresponding project records.

Testing will focus on the accuracy of project-image associations, date information, and the presentation of available observations.

Sprint 4: System Integration

The fourth sprint will focus on integrating the major components of MASID into a unified prototype. Procurement information, project records, geographic visualization, and satellite-based information will be connected through the project identifier and associated database relationships.

The researchers will implement the integrated project view and temporal comparison functions. Integration testing will be conducted to identify inconsistencies between system components.

Sprint 5: Refinement and Evaluation Preparation

The fifth sprint will focus on refining the integrated prototype. Functional errors, interface inconsistencies, data presentation issues, and integration problems identified during previous iterations will be addressed.

The researchers will review MASID against the established functional and non-functional requirements and prepare the system for formal assessment. Testing will be repeated after major corrections to verify that the implemented changes operate as intended.

Proposed Technology Stack

The following technologies are proposed for the implementation of MASID. The final technologies and versions shall be confirmed based on the actual development environment and requirements of the researchers.

Table 5: *Proposed Tools and Technologies Used in Development*

Tool / Technology	Proposed Purpose
Visual Studio Code	Integrated development environment for coding and debugging
Git	Version control

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Tool / Technology	Proposed Purpose
GitHub	Source-code management, collaboration, and version tracking
Python	Data processing, geospatial processing, and satellite-data analysis
Jupyter Notebook	Exploratory data analysis and preprocessing development
Pandas	Tabular procurement and project-data manipulation
NumPy	Numerical computation
GeoPandas	Geospatial data manipulation and analysis
Shapely	Geometric operations
Rasterio	Raster and satellite-image processing
GDAL	Geospatial data conversion and raster/vector processing
Scikit-image	Image-processing operations where required
Matplotlib	Exploratory visualization and analytical plots
FastAPI	Backend API and processing services
PostgreSQL	Relational database management
PostGIS	Spatial database storage and geographic queries
Next.js	Web application framework
React	Frontend component development
TypeScript	Typed frontend/application development
Tailwind CSS	User-interface styling
Leaflet	Interactive geographic mapping
QGIS	Geospatial inspection, validation, and preparation
Postman	API testing
Chrome DevTools	Browser debugging and performance inspection
Vercel / equivalent	Web application deployment, subject to final hosting decision

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Table 5 – continued from previous page

Tool / Technology	Proposed Purpose
Cloud/backend hosting service	Backend/API deployment, subject to final infrastructure decision

Data Processing Technologies

Python will serve as the primary computational environment for data preparation and geospatial processing. Pandas will be used to manipulate structured procurement and project datasets, while GeoPandas and Shapely will support geographic data operations. Rasterio and GDAL may be used for raster and satellite-image processing where required by the selected imagery and analytical procedures.

Jupyter Notebook may be used during exploratory analysis and preprocessing to document data inspection, transformation, and validation procedures before finalized processing routines are incorporated into the system.

Table 6: *Proposed Python Libraries*

Python Package	Primary Purpose in MASID
Pandas	Cleaning, transforming, merging, and organizing procurement and project datasets
NumPy	Numerical operations and array-based computation, particularly for image and geospatial data processing
GeoPandas	Manipulation and analysis of geospatial vector data such as project coordinates, boundaries, and shapefiles
Shapely	Geometric operations involving project points, lines, polygons, buffers, and spatial relationships
Rasterio	Reading, writing, transforming, and processing raster-based satellite imagery

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Table 6 – continued from previous page

Python Package	Primary Purpose in MASID
GDAL	Geospatial data conversion, coordinate reference system handling, raster/vector processing, and interoperability
OpenCV	Image processing and computer-vision operations on satellite imagery
scikit-image	Image enhancement, segmentation, feature extraction, and other image-processing operations, if required
Matplotlib	Generation of charts, plots, and visualizations for exploratory and analytical purposes
Seaborn	Statistical visualization and exploratory analysis of procurement and project-related datasets, if used
Scikit-learn	Machine-learning preprocessing, feature extraction, model development, and evaluation, if an ML component is included
PyProj	Coordinate reference system transformation and geographic coordinate conversion
Requests	Retrieval of data from permitted APIs or web-based data services, if applicable
Pydantic	Data validation and structured data handling for backend/API inputs, if using FastAPI
FastAPI	Development of Python-based backend/API services for computational or image-processing tasks, if needed
Uvicorn	Running the FastAPI application server, if FastAPI is used

The final library set will depend on the satellite data format and analytical method ultimately adopted by the researchers.

Frontend Architecture

The frontend of MASID will provide the primary interface through which users interact with the system. A component-based web framework will be used to organize the interface into reusable components. The frontend will include the interactive map, project markers, project information panels, procurement information, satellite visualization, filtering controls, and temporal comparison interface.

The interface will communicate with the backend through defined application interfaces to retrieve project records, geographic information, procurement information, and available satellite observations.

Backend Architecture

The backend will handle application logic, data retrieval, data association, and communication between the frontend and database and processing components. A Python-based API service such as FastAPI may be used for computational and data-processing operations, particularly for geospatial and satellite-related processing.

Backend functions will validate incoming requests, retrieve the required records, process information where necessary, and return structured results to the frontend.

Database Implementation

PostgreSQL will be used as the proposed relational database management system, with PostGIS providing spatial database capabilities where required. The database will store project records, procurement information, geographic attributes, satellite metadata, and other system information.

Spatial database functionality will allow geographic records to be queried and associated with project records based on their stored geometry.

GIS Implementation

Leaflet will be used as the proposed interactive mapping library for displaying flood control project locations. Geographic data prepared through Python and/or QGIS will be

transformed into formats suitable for web-based visualization.

The map will allow users to navigate geographic areas, identify project markers, and retrieve project information associated with individual locations.

Satellite Data Processing

Satellite imagery will be processed using the selected image-processing and geospatial libraries. Depending on the final data source and imagery characteristics, preprocessing may include image selection, metadata inspection, cloud screening, geographic clipping, coordinate alignment, and preparation for visualization or analysis.

The specific satellite-processing procedure will be determined after the researchers establish the final satellite data source and analytical requirements.

Procurement Data Integration

Procurement records will be integrated with project records through common identifiers or other validated attributes. Where direct identifiers are unavailable, project associations will only be established using documented and defensible matching procedures.

The system will retain the source attributes of procurement records to maintain traceability between the displayed information and the underlying dataset.

Security and Data Integrity

System security and data integrity will be considered during implementation. Backend operations will validate incoming data and requests before processing or storage. Database constraints and structured data models will be used to reduce invalid records and maintain relationships between project, procurement, geographic, and satellite information.

Source data will be preserved separately from processed datasets where practical to allow the researchers to trace transformations performed during data preparation. Changes made during data cleaning and transformation will be documented to maintain reproducibility.

Testing and Debugging

Testing and debugging will be conducted throughout the Agile development process. Rather than postponing testing until the completion of development, individual functions and components will be tested during their respective sprints.

Testing will determine whether MASID operates according to its functional requirements and whether different system components communicate correctly. Identified defects will be documented, investigated, corrected, and retested.

Table 7: MASID Testing Matrix

Testing Area	Purpose	Example Test
Functional Testing	Verify individual system functions	Selecting a project retrieves the correct project record
Integration Testing	Verify interaction between components	Project selection retrieves procurement and satellite information
Database Testing	Verify data storage and retrieval	Correct project record is returned for a valid project ID
Geospatial Testing	Verify geographic accuracy	Project marker corresponds to the prepared project coordinates
Satellite Testing	Verify image/project association	Selected project retrieves the correct available observation
Temporal Testing	Verify date-based comparison	User can navigate between available observation periods
UI Testing	Verify interface behavior	Controls and navigation operate as intended
Performance Testing	Assess system response	Map and project records load within acceptable test conditions
Compatibility Testing	Assess browser compatibility	System operates on supported modern browsers
Data Validation	Identify data inconsistencies	Missing or malformed records are detected

Functional Testing

Functional testing will verify whether the implemented features operate according to the established requirements. Tests will cover project search, project selection, map visualization, retrieval of procurement records, retrieval of satellite observations, filtering, temporal

comparison, and navigation.

Table 8: *Sample Functional Test Cases*

Test ID	Function	Input	Expected Result
TC-01	Project Search	Valid project keyword	Matching project records displayed
TC-02	Project Selection	Project marker	Correct project information displayed
TC-03	Procurement Retrieval	Valid project ID	Associated procurement information displayed
TC-04	Map Display	Valid geographic record	Correct project marker displayed
TC-05	Satellite Retrieval	Valid project/date	Available satellite observation displayed
TC-06	Temporal Comparison	Two available dates	Corresponding observations displayed
TC-07	Filtering	Valid filter criterion	Matching projects displayed
TC-08	Invalid Record	Invalid project ID	Appropriate no-result/error response

Geospatial Validation

Geospatial validation will assess whether project records are represented at the appropriate geographic locations based on the prepared source information. Coordinates will be checked for validity, consistency, and correspondence with the associated project records.

Where geographic uncertainty exists, the uncertainty will be documented rather than concealed. The researchers will distinguish between a software error in geographic rendering and an uncertainty or limitation originating from the source dataset.

Satellite Data Validation

Satellite-data validation will examine whether the imagery presented by MASID corresponds to the intended project location and observation period. Image metadata and geographic boundaries will be inspected to verify that the imagery is appropriately associated with the selected project.

The researchers will also document satellite-data limitations such as unavailable observations, cloud coverage, spatial resolution, and temporal gaps.

Deployment

The deployment phase involves transitioning MASID from the development environment into a controlled research environment for demonstration and evaluation. The researchers will configure the frontend, backend, database, processing services, and prepared datasets required for the operation of the prototype.

Prior to deployment, the researchers will conduct final integration checks to verify that system components communicate correctly. The deployed prototype will then be demonstrated to the identified evaluators as part of the system assessment.

Deployment in this study is limited to the research prototype and evaluation environment. It does not constitute nationwide operational deployment, continuous monitoring, or formal institutional adoption by DPWH or other government agencies.

Review

The Review stage will involve examining the results of each Agile iteration against the requirements and objectives established during the planning and requirements phases. The researchers will review functional performance, data integration, geographic visualization, satellite-data presentation, interface usability, and identified technical limitations.

Issues discovered during testing will be documented and prioritized. Feedback obtained from thesis advisers and identified evaluators will be considered in determining whether further refinements are necessary. The results of each review will guide succeeding development activities and ensure that the final prototype remains aligned with the objectives and scope of

the study.

System Evaluation

Upon completion of the iterative refinement process undertaken through the Agile development cycle, the study will proceed to a formal evaluation of the MASID prototype to determine the extent to which the developed system satisfies its established functional and non-functional requirements. Whereas the testing activities conducted within individual sprints are primarily concerned with verifying technical correctness at the component level, the system evaluation constitutes a distinct and more comprehensive undertaking, intended to elicit a structured, criterion-based assessment of the prototype's overall quality from qualified evaluators external to the development process. This distinction is methodologically significant, as it ensures that the assessment of MASID extends beyond internal developer verification and incorporates independent, informed judgment regarding the system's usability, functional adequacy, and fitness for its intended institutional purpose.

Evaluation Framework

The evaluation of the MASID system will follow the ISO/IEC 25010 Software Product Quality Model, which is a popular approach for evaluating software systems on multiple statistically evidenced quality attributes. Out of the eight listed in this model, only those to which the present study is highly relevant to its system objectives, as expressed by the system's non-functional requirements as described above in Table 3, will be explored.

Functional Suitability – the fulfillment of the functions embodied in MASID with respect to the requirements determined in the requirements analysis, including project visualization, extraction of procurement information, presentation of satellite data, and comparative analysis in terms of time.

Performance Efficiency – reflects the system's responsiveness to loading and displaying maps, project information, and satellite imagery in a particular test environment.

Compatibility – confirms that the web application is compatible with multiple modern browsers.

Usability – a measure of how well the users are able to maneuver through the interface, find their required projects, and understand presented information with little or no effort.

Reliability – the ability of the system to produce the same results given the same valid data and operations.

Security – the degree to which the system's access controls are adequate to protect resources and information stored within it.

Maintainability – the extent to which the components of the system are comprehensible and to which their structure allows for their easy modification.

Data Integrity – the extent to which the information provided by MASID maintains the relationships and values that existed in prepared datasets.

Each characteristic will be operationalized into evaluation statements that will be rated by the identified evaluators using the instrument described in a subsequent section.

Evaluators of the Study

The respondents for the system evaluation will be selected using purposive sampling based on their expertise and stake in the developed system. There are two categories of evaluators that will be formed for this study:

1. **Technical Evaluators** consisting of the thesis adviser, the thesis consultant, and others with sufficient background in software development, geographic information systems, or data science who will evaluate the more technical aspects of the system, namely the functional suitability, performance efficiency, reliability, security, maintainability, and data integrity of MASID.
2. **Domain Evaluators** consist of relevant officials from the Department of Public Works and Highways (DPWH) and, if available, the Philippine Space Agency (PhilSA) who will evaluate the practicality and relevance of MASID from the perspective of its intended end users.

The rationale for having two categories of evaluators is to ensure that the system will be assessed both from a theoretical and practical perspective.

Evaluation Instrument

A questionnaire constructed by the researcher based on the ISO/IEC 25010 quality characteristics described above will be used to collect data. The questionnaire will contain descriptive statements related to each characteristic that will have to be rated using the five-point Likert scale shown below:

Table 9: *Five-Point Likert Scale*

Scale	Range	Descriptive Rating
5	4.20–5.00	Strongly Agree / Excellent
4	3.40–4.19	Agree / Very Satisfactory
3	2.60–3.39	Neutral / Satisfactory
2	1.80–2.59	Disagree / Fair
1	1.00–1.79	Strongly Disagree / Poor

Prior to administration, the instrument will be subjected to review by the thesis adviser and consultant, in order to establish its content validity and to ascertain its appropriateness for the intended respondents. Refinements to the questionnaire will be undertaken as warranted by the outcomes of this review.

Data Gathering Procedure

Evaluation will take place after the deployment of the MASID prototype in the research and evaluation environment described in the previous Deployment section. A live demonstration of the system will take place for each evaluator or groups of evaluators, during which the key functionalities of MASID, including the project visualization, procurement information retrieval, satellite data presentation, and temporal comparison will be demonstrated. After the short demonstration, the evaluators will be given an opportunity to directly interact with the system (where applicable) before the completion of the evaluation questionnaire.

Responses will be gathered either with a printed instrument or the equivalent digital form, depending on the availability and preference of the respective evaluators. Researchers

will make sure that evaluators are given enough time and background to make an informed assessment of the system.

Statistical Treatment of Data

Responses gathered through the evaluation instrument will be tabulated and analyzed using descriptive statistics. A weighted mean will be calculated for each quality characteristic, as well as the overall system evaluation, using the following formula:

$$WM = \frac{\sum fx}{N}$$

where WM stands for the weighted mean, f is the frequency of each response according to the scale points, x is the value of the scale point, and N is the total number of respondents. The weighted means will be interpreted using the descriptive rating scale provided above to determine the overall level of acceptability of MASID among the surveyed groups of evaluators.

Qualitative comments and observations provided by the evaluators will also be collected and thematically analyzed, where applicable, to support the quantitative results and inform potential recommendations for the future development or improvement of the system.

Ethical Considerations

The study will observe appropriate ethical considerations throughout the collection, processing, analysis, and presentation of data. The researchers shall use the government's procurement and project records for academic and research purposes only. Information that is not necessary for the objectives of the study will not be collected, processed, or disclosed. Where access to restricted government records or satellite datasets requires authorization, the researchers will comply with the applicable data-use agreements, institutional requirements, and conditions established by the data providers. The researchers will also ensure that satellite imagery and other geospatial data are used only within the permitted scope of their respective licenses and agreements. The team has identified academic and non-commercial use as the intended basis for requested PhilSA data, together with the necessary university endorsement and data-use agreements.

Responsible Interpretation of Results

The interpretation of MASID results will be conducted responsibly. Satellite-based discrepancies will be treated as indicators for further verification and will not, by themselves, be interpreted as proof of fraud, corruption, legal noncompliance, or project failure. This is important because satellite imagery and Machine Learning outputs may be affected by factors such as cloud cover, vegetation, surrounding structures, image resolution, temporal availability, and the accuracy of project coordinates. The results will therefore be presented with consideration of the limitations of the data and methods used in the study.

Data Privacy and Confidentiality

The privacy and confidentiality of the data and information of individuals who may participate in the system evaluation and field validation activities will be respected. Any personal information obtained during the research will be kept strictly confidential and will not be disclosed without appropriate consent or authorization.

Academic Integrity

The researchers will uphold academic integrity by presenting accurate and precise information regarding the procedures, outcomes, limitations, and sources of the study. The data and system outputs will not be falsified, manipulated, or misrepresented to achieve a desired result. All government records, satellite data, references, and other materials used for the study will be cited appropriately and acknowledged.